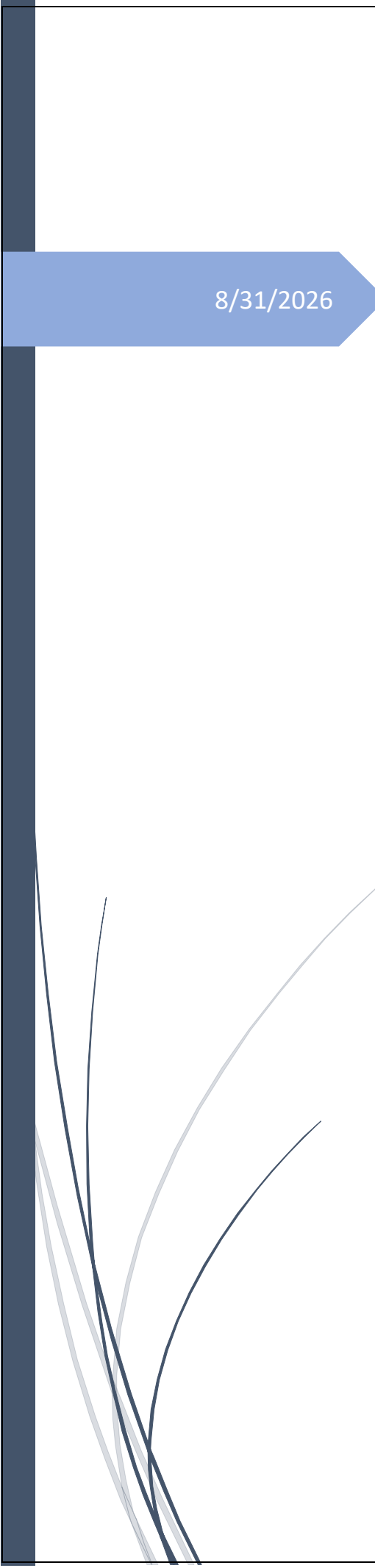


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8/31/2026

# Software Requirement Specification

**Wellness App & WebApp for Student Mental Health**

WORK INTEGRATED LEARNING IN INFO TECHNOLOGY  
Central University of Technology

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## Group members

| Group F          |                |
|------------------|----------------|
| Name and Surname | Student Number |
| BF Motseki       | 223038085      |
| Lesedi D Mokheti | 223051025      |
| ML Mwenda        | 221034577      |
| A Jara           | 223007530      |
| TV Thabisi       | 223005893      |
| S Bila           | 223048044      |
| B Mbinga         | 223020021      |

## Stakeholder and organisation

### Organisation name and type

CUT Wellness Centre , University Wellness and Student Support Centre

| Name        | Role        | Contact details   |
|-------------|-------------|-------------------|
| C.Maluka    | Stakeholder | cmaluka@cut.ac.za |
| Dr N.Afrika | Stakeholder | nafrika@cut.ac.za |

### Contact person

Cleo Maluka – Senior Unit Administrator, Wellness Centre

### Organisation's main activities

The Wellness Centre provides psychological counselling, health advice, medication related services within its scope of practice, and referrals to hospitals or specialists when required.

### Reason for selecting the stakeholder.

The stakeholder was selected because they have professional knowledge of the Wellness Centre's services and processes and can provide valuable information about the challenges students experience when accessing wellness services. Their input helps the group understand the current appointment-booking process and identify requirements for improving the way students book appointments and how practitioners manage their availability and appointments.

## Problem Statement

Many students face significant mental health challenges, stress and academic pressure. Cut wellness centre prioritises students health and psychological needs they provide counselling services. According to the Wellness centre, student mental health concerns have increased substantially in recent years, yet many students who need support do not access it

Many students go to the wellness centre to book appointments to seek help. The organisation currently uses paper diaries and manual communication to complete student mental health service management. The administration has no digital tools to keep track students progress. Students seeking help face stigma barriers, making it difficult to visit in person. Students have **no digital platform** for self-monitoring ,self-help, appointment reminders. The communication between the students and service providers Is limited to in-person. This results in duplicated bookings and scheduling errors, high no show rates. Inefficient use of therapist and admin time. Delayed support for students in crisis. No early warning system for at-risk students

### 1. Purpose

CampusCare aims to digitally transform mental health support services that the Central University of Technology by replacing manual, paper-based processes with an integrated mobile and web platform. Purpose is to support students through their academic journey. To provide students with **immediate, accessible, and private mental health support** through self-monitoring tools, peer connection, and emergency response, while empowering wellness staff with the visibility needed to **identify and support at-risk students early and offer post consultation support**—all within a secure digital platform.

### 2. Definitions and abbreviations

| Term or abbreviation | Meaning                                          |
|----------------------|--------------------------------------------------|
| <b>SRS</b>           | Software Requirements Specification              |
| <b>Flutter</b>       | Framework used to develop the mobile application |
| <b>ASP.NET</b>       | Framework used to develop the web application    |
| <b>Supabase</b>      | Shared online backend and database               |

## Project scope

### **Project Title:**

#### **CampusCare**

CUT Student wellness appointment system

A mobile and web-based platform designed to provide university students with accessible mental health and wellness support.

### **Aim:**

To design, develop, and deploy a secure, POPIA-compliant mobile and web platform that provides university students with accessible mental health support tools and connects them with CUT professional help when needed.

### **Objective:**

- Enable students to track their mood and emotional wellbeing.
- Provide stress-management and mindfulness resources.
- Provide journaling and emotional check-in features.
- Provide medication reminders.
- Encourage engagement through points, streaks and rewards.
- Enable secure communication and appointment scheduling with therapists/psychologists.
- Provide emergency support features.
- Protect student information through POPIA-compliant data processing.

### **Deliverable:**

- Functional mobile application for iOS and Android.
- Functional web application with administrative features.
- UI/UX designs and prototypes.
- Database schema and ERD.
- API documentation.
- Integration testing and QA report.
- Security audit and POPIA compliance report.

- Technical documentation and user manuals.
- Final project presentation and demonstration.

### **Acceptance Criterion:**

1. Both the mobile and web applications are successfully deployed.
2. The core wellness features are functional.
3. Secure therapist/student information
4. POPIA compliance has been verified.
5. Positive feedback is received from a pilot group of university students.
6. The final project is successfully demonstrated and defended.
7. Technical and user documentation has been completed.

### **Exclusion:**

The system will not integrate with existing university systems.

### **Constraint:**

- The project must be completed within 9 weeks.
- The budget is limited to free/open-source tools and services.
- There will be no integration with existing university systems.
- The mobile application must support both iOS and Android.

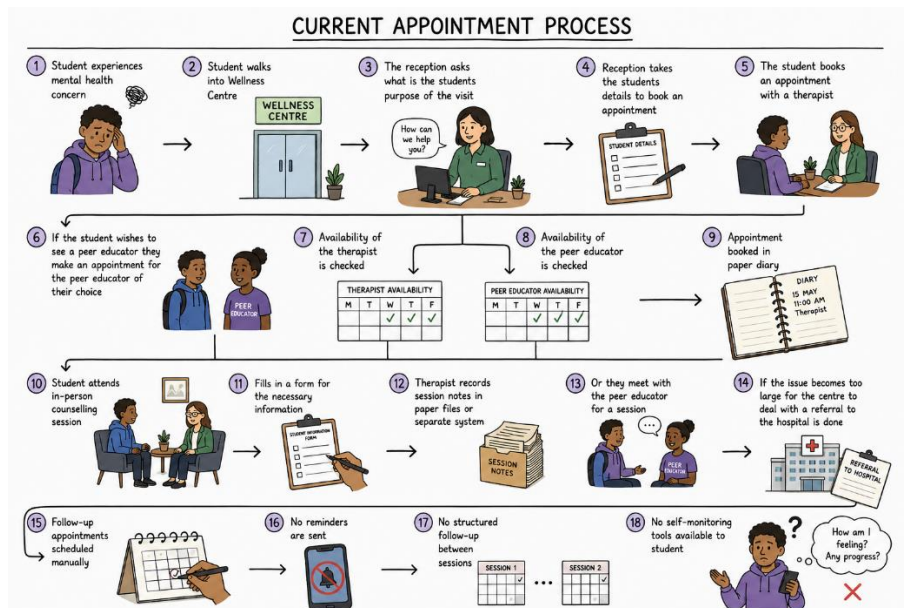
### **Assumption:**

- Students will use their university email address for verification.
- Therapists/psychologists will be pre-registered by the university.
- Students will have access to a smartphone or computer with internet connectivity.

## stake holder and user groups

| Stakeholder or user            | Role in the proposed system                       |
|--------------------------------|---------------------------------------------------|
| <b>External Stakeholder</b>    | Cleo Maluka                                       |
| <b>Mobile application user</b> | Students                                          |
| <b>Web applications user</b>   | Therapist, peer educators, receptionist, students |

## Current Process



1. Student experiences mental health concern
2. walks into Wellness Centre
3. The reception asks what is the students purpose of the visit
4. Reception takes the students details to book an appointment
5. The student books an appointment with a therapist
6. If the student wishes to see a peer educator they make an appointment for the peer educator of their choice
7. Availability of the therapist and peer educator is checked
8. Appointment booked in paper diary
9. Student attends in-person counselling session
10. Fills in a form for the necessary information
11. Therapist records session notes in paper files or separate system
12. Or they meet with the peer educator for a session
13. If the issue becomes too large for the centre to deal with a referral to the hospital is done
14. Follow-up appointments scheduled manually
15. No reminders are sent
16. No structured follow-up between sessions

## 17. No self-monitoring tools available to student

### Product perspective

The Student Wellness System is an **integrated digital platform** consisting of three interconnected components:

- **Flutter Mobile Application** – Student-facing app for mood tracking, journaling, appointment booking, and wellness resources.
- **[ASP.NET](#) Web Application** – Staff-facing dashboard for managing appointments, viewing student data, and identifying at-risk students.
- **Supabase Backend** – Shared database, authentication, real-time synchronisation, and file storage.

All three components operate as **one unified system** through the Supabase backend. Information entered through either platform is stored in the same database and synchronised instantly. This ensures:

- An appointment booked on Flutter appears on [ASP.NET](#)
- A mood entry logged on Flutter is visible to the therapist on [ASP.NET](#)
- Practitioner verification status is consistent across both platforms
- Student wellness data remains identical across both applications

### Product functions

| User Role             | Functions                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Students</b>       | Register/login, complete digital intake form, capture personal and next-of-kin information, provide consent, book/view/cancel appointments, log moods with streak history, complete emotional check-ins, create journal entries, access wellness resources, set medication reminders, use emergency SOS, and manage their profile, access to community resources |
| <b>Therapist</b>      | Log in, view verification status, manage availability, view appointments, view authorised student information (moods, journals, check-ins), and manage therapist records                                                                                                                                                                                         |
| <b>Administrators</b> | Manage users, manage and verify Therapist, manage appointments, view system reports, and access analytics                                                                                                                                                                                                                                                        |
| <b>Peer Educators</b> | Log in, view verification status, manage availability, view and manage student bookings, view authorised student information (limited), send                                                                                                                                                                                                                     |

|  |                                                                                                    |
|--|----------------------------------------------------------------------------------------------------|
|  | and receive secure messages with assigned students, manage own profile, and view health broadcasts |
|--|----------------------------------------------------------------------------------------------------|

## Functional Requirement, Non-functional, Integration, Prioritisation

Requirement Table

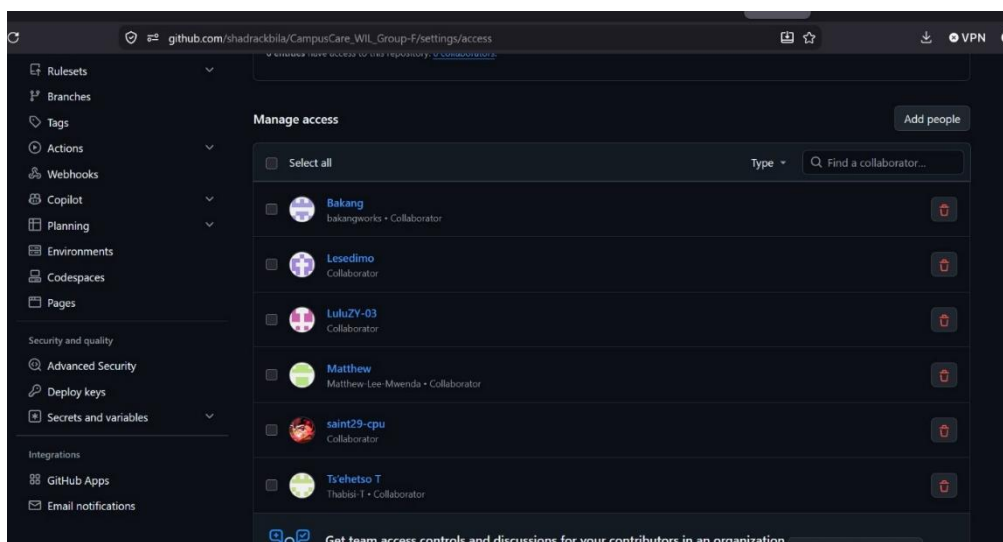
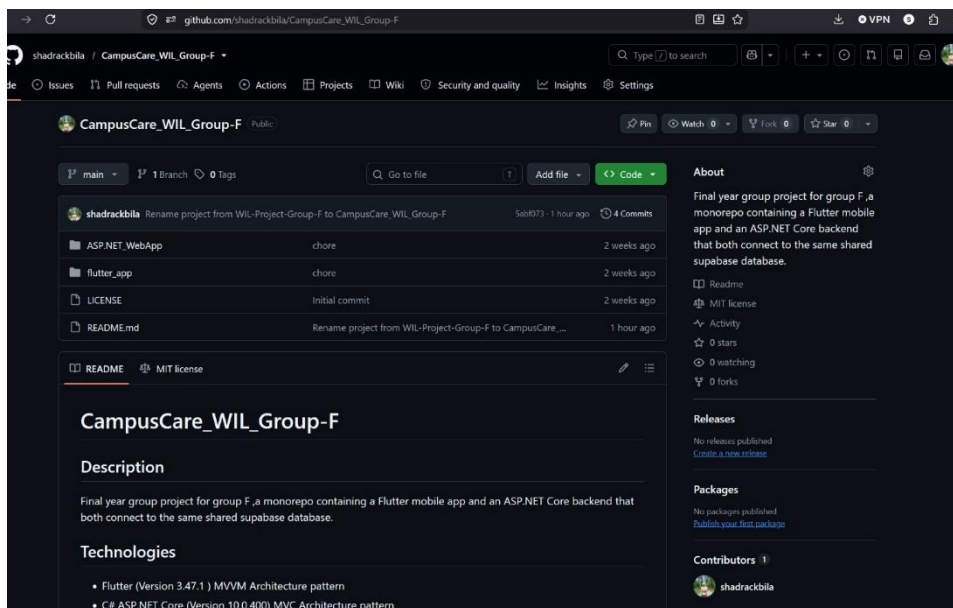
| ID   | REQUIREMENT                                                                                                 | TYPE           | PRIORITY  |
|------|-------------------------------------------------------------------------------------------------------------|----------------|-----------|
| FR1  | The student must be able to register and log in.                                                            | Functional     | Must Have |
| FR2  | The student must be able to complete the Wellness Centre Intake Form.                                       | Functional     | Must Have |
| FR3  | The student must be able to provide next-of-kin information.                                                | Functional     | Must Have |
| FR4  | The student must be able to select the purpose of the visit.                                                | Functional     | Must Have |
| FR5  | The student must be able to provide consent.                                                                | Functional     | Must Have |
| FR6  | The student must be able to submit an appointment request.                                                  | Functional     | Must Have |
| FR7  | The student must be able to view and manage their appointments.                                             | Functional     | Must Have |
| FR8  | The practitioner must be able to manage their availability.                                                 | Functional     | Must Have |
| FR9  | The practitioner must be able to view authorized student information.                                       | Functional     | Must Have |
| FR10 | The administrator must be able to verify practitioners.                                                     | Functional     | Must Have |
| FR11 | The student must be able to create, update, and delete their own mood entries.                              | Functional     | Must Have |
| FR12 | The student must be able to create, update, and delete their own journal entries.                           | Functional     | Must Have |
| FR13 | The student must be able to manage their own emotional check-ins.                                           | Functional     | Must Have |
| NFR1 | Only authorized users may access protected student information.                                             | Non-functional | Must Have |
| NFR2 | Practitioners must only be able to view authorized student wellness records.                                | Non-functional | Must Have |
| NFR3 | Practitioners must not be able to create, update, or delete student-owned wellness records.                 | Non-functional | Must Have |
| SR1  | The mobile application must be developed using Flutter.                                                     | System         | Must Have |
| SR2  | The web application must be developed using ASP.NET/C#.                                                     | System         | Must Have |
| IR1  | Information submitted through Flutter must be stored in Supabase and available in ASP.NET where authorized. | Integration    | Must Have |
| IR2  | Information submitted through ASP.NET must be stored in Supabase and available in Flutter where authorized. | Integration    | Must Have |

|      |                                                                                                             |             |             |
|------|-------------------------------------------------------------------------------------------------------------|-------------|-------------|
| IR3  | Appointment updates made through either platform must be reflected on the other platform.                   | Integration | Must Have   |
| IR4  | Student wellness data permissions must be enforced consistently across both platforms.                      | Integration | Must Have   |
| FR14 | The student should be able to use medication reminders.                                                     | Functional  | Should Have |
| FR15 | The system should provide streaks and achievements.                                                         | Functional  | Should Have |
| FR16 | The system should provide additional reporting.                                                             | Functional  | Should Have |
| IR1  | Information submitted through Flutter must be stored in Supabase and available in ASP.NET where authorized. | Integration | Must Have   |
| IR2  | Information submitted through ASP.NET must be stored in Supabase and available in Flutter where authorized. | Integration | Must Have   |

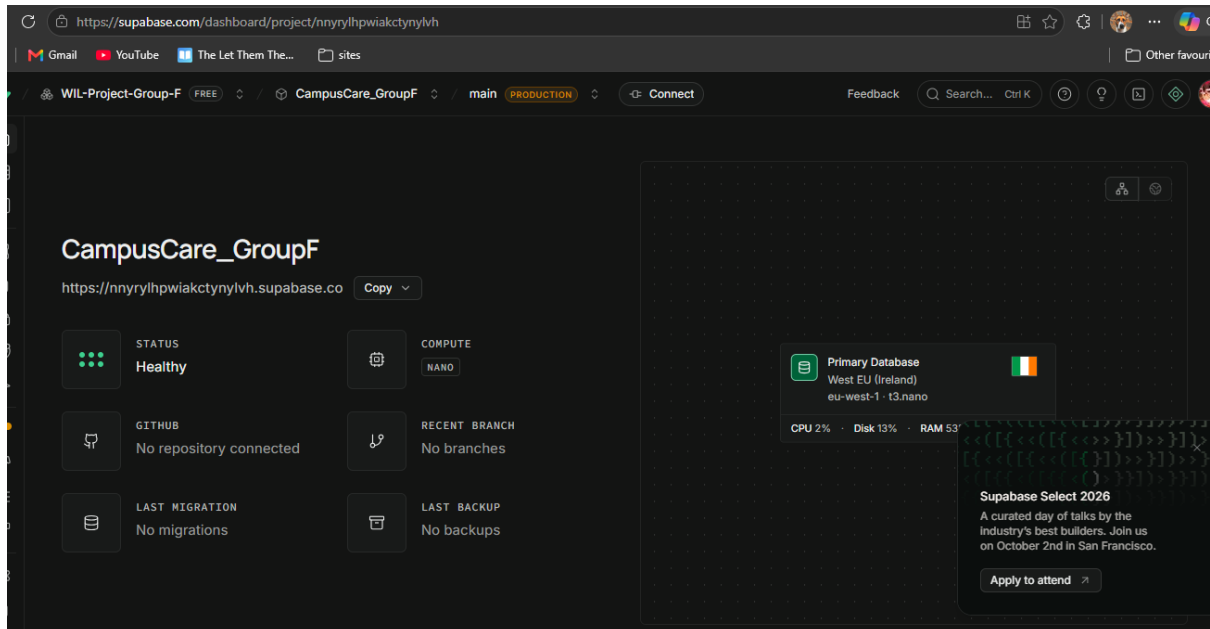
## Project Setup

### 2.1 GIT HUB

Git Hub Link : [https://github.com/shadrackbila/CampusCare\\_WIL\\_Group-F.git](https://github.com/shadrackbila/CampusCare_WIL_Group-F.git)



## 2.2 Supabase Setup



## Stakeholder Evidence

| Stakeholder Interview       |                                                                                           |              |               |
|-----------------------------|-------------------------------------------------------------------------------------------|--------------|---------------|
| Date :                      | 20 August 2026                                                                            | Time         | 12:05 - 12:25 |
| Location                    | CUT Wellness Centre offices                                                               |              |               |
| Role                        | Responsibility                                                                            | Group Member |               |
| <b>Facilitator</b>          | Leads the interview                                                                       | A Jara       |               |
| <b>Note-taker</b>           | Records responses                                                                         | TV Thabisi   |               |
| <b>Timekeeper</b>           | Monitors the meeting                                                                      | LD Mokheti   |               |
| <b>Requirements Checker</b> | Summarizes and confirms the needs                                                         | BF Motseki   |               |
| <b>Observer</b>             | Observes stakeholder body language and non-verbal cues; notes unspoken concerns           | S Bila       |               |
| <b>Technical Advisor</b>    | Explains technical concepts in simple terms; answers technical questions about the system | ML Mwenda    |               |
| <b>Stakeholder Liaison</b>  | follows up after the meeting                                                              | B Bakang     |               |

## Interview Questions and Stakeholder responses

| Question                                               | Stakeholder Response                                                                                                                                                                                                       | Follow-up Required | Confirmed? |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------|
| How does the current appointment booking process work? | Students currently book appointments by physically visiting the Wellness Centre. The process is largely manual.                                                                                                            | No                 | Yes        |
| 1. What services does the Wellness Centre provide?     | Psychological counselling, Health advice. Medication-related services within the center's scope of practice. Referrals to hospitals or specialists when a student's condition falls outside the Centre's scope of practice | Yes                | Yes        |

|                                                             |                                                                                                                                                                                                                                                                                                                             |                                      |     |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----|
| 2. What problems or challenges are currently experienced?   | The manual process is an inconvenience for students. As they must physically book an appointment which poses as an inconvenience on days they are not on Campus and must travel to simply book an appointment. Another is that students do not have a centralized way to see which psychologists/counselors are available.  | No                                   | Yes |
| 3. What information is required when making an appointment? | Students provide information such as their name, student number, and contact details. There are also confidentiality and consent requirements that need to be communicated to students before receiving certain services.                                                                                                   | No                                   | Yes |
| 4. What services should the proposed system support?        | The Wellness Centre provides both psychological counselling services and healthcare services. Further consultation with the clinic general manager will be required to understand their specific challenges and to determine whether the services should be incorporated into one system or treated as separate components. | Yes                                  | Yes |
| 5. What information should students see before booking?     | Students should be able to see the available practitioners and appointment slots, including practitioner availability. And should be able to select the preferred gender of the practitioner.                                                                                                                               | No                                   | Yes |
| 6. What security requirements are important?                | Patient confidentiality and protection of sensitive information are critical. Real patient data should not be accessible to the development team                                                                                                                                                                            | Yes – Consult with CUT IT Department | Yes |
| 7. Do you have any specific expectations from the project?  | The stakeholder explained CUT's goal is to move from paper-based processes towards becoming a paperless institution by 2030. Therefore, the project is expected to support this broader technological transformation.<br>The key expectation is that the system should digitize the appointment booking                     | No                                   | Yes |

|                                                                   |                                                                                                                                                                                                                                          |     |     |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
|                                                                   | process while improving convenience and accessibility for students.                                                                                                                                                                      |     |     |
| 8. What do you think would derail this project?                   | A breach of patient confidentiality or violation of institutional privacy policies could significantly affect or derail the project.<br>Technical glitches, system failure or security problems could also negatively affect the project | Yes | Yes |
| 9. Would you be willing to participate in future project reviews? | Yes. The stakeholder is willing to review the proposed solution and provide feedback, advice, corrections, and additional requirements.                                                                                                  | No  | Yes |
| 10. How should we communicate with you for project reviews?       | The stakeholder communicated that they can be contacted through: <ul style="list-style-type: none"> <li>• Direct phone call</li> <li>• WhatsApp</li> </ul>                                                                               | No  | Yes |

| Stakeholder Follow up meeting |                                                                                           |              |               |
|-------------------------------|-------------------------------------------------------------------------------------------|--------------|---------------|
| Date :                        | 26 August 2026                                                                            | Time         | 17:01 – 17:20 |
| Location                      | Teams Meeting : Follow Up meeting                                                         |              |               |
| Role                          | Responsibility                                                                            | Group Member |               |
| <b>Facilitator</b>            | Leads the interview                                                                       | A Jara       |               |
| <b>Note-taker</b>             | Records responses                                                                         | TV Thabisi   |               |
| <b>Timekeeper</b>             | Monitors the meeting                                                                      | LD Mokheti   |               |
| <b>Requirements Checker</b>   | Summarizes and confirms the needs                                                         | BF Motseki   |               |
| <b>Observer</b>               | Observes stakeholder body language and non-verbal cues; notes unspoken concerns           | S Bila       |               |
| <b>Technical Advisor</b>      | Explains technical concepts in simple terms; answers technical questions about the system | ML Mwenda    |               |
| Stakeholder Liaison           | follows up after the meeting                                                              | B Bakang     |               |

## Meeting Evidence

### Stakeholder Project Review And Discussion

Wed Aug 26, 2026

19:00–20:00

Microsoft Teams Meeting

 [Share meeting invite](#)

Join

Edit

Microsoft Teams meeting Join: <https://teams.microsoft.com/meet/350365753886072?p=QipElsigoMwE92seOD> Meeting ID: 350 365

[See more](#)

## Meeting Recap



Wednesday Aug 26, 2026

19:01 - 19:19

 2 transcripts

[View recap](#)



Meeting options



Cancel event

## Feasibility Study

### 2.3 Technical Feasibility

#### Technical Requirements

The proposed Student Wellness System will consist of a Flutter mobile application and an ASP.NET/C# web application, with both platforms using Supabase as the shared backend/database. The system will allow information submitted through either platform to be stored and accessed by authorised users across both platforms. The technical feasibility assessment considered the group's available skills, development technologies, hardware, testing devices, internet connectivity and the complexity of the proposed functionality.

#### Technical feasibility Findings

| Area                   | Finding                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Flutter development    | The group will use Flutter for the mobile application.                                                                                              |
| ASP.NET/C# development | The web application will be developed using ASP.NET/C#.                                                                                             |
| Supabase               | Supabase will be used as the shared database/backend.                                                                                               |
| System integration     | Flutter and ASP.NET will connect to the same Supabase backend                                                                                       |
| Development hardware   | Development computers are required for both Flutter and ASP.NET development.                                                                        |
| Testing devices        | Smartphones and computers will be required to test the applications.                                                                                |
| Internet connectivity  | Internet access is required for development, Supabase and stakeholder communication.                                                                |
| Skills development     | The group may need to strengthen skills in Supabase integration, authentication, role-based access and secure data handling.                        |
| Technical challenges   | Secure communication, emergency support, role-based access and synchronisation between the two platforms may require additional development effort. |

#### Technical Feasibility Decision

##### Decision: Feasible with changes

The system is technically feasible because the required technologies have been identified and the Flutter and ASP.NET applications can use the same Supabase

backend. However, some functions may be technically challenging for the group and may require additional learning, testing and prioritisation.

## 2.4 Operational Feasibility

The proposed system is intended to address the current manual appointment-booking process at the CUT Wellness Centre. Currently, students must physically visit the Wellness Centre to make appointments, and there is no centralised digital platform for viewing practitioner availability and appointment slots. In addition, practitioners do not have an efficient digital system to manage their clients, appointments, availability, and relevant student information, resulting in greater reliance on manual processes.

The proposed system will support students, practitioners and administrators, with different functionality based on their roles

### Intended Users

| User                  | Main Responsibilities/Funtions                                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Students</b>       | Register, complete intake forms, provide consent, book and manage appointments, complete wellness activities and manage their profiles.                   |
| <b>Practitioners</b>  | Manage their availability, view and manage appointments, access authorised student information, and manage relevant practitioner and session information. |
| <b>Administrators</b> | Manage users, practitioners, practitioner verification, appointments and reports.                                                                         |

## Operational Feasibility Findings

| Area                         | Finding                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Problem addressed</b>     | The system will digitise the manual appointment-booking process.                                             |
| <b>User suitability</b>      | Students, practitioners and administrators are the intended users.                                           |
| <b>Device requirements</b>   | Students, practitioners and administrators are expected to access the system using smartphones or computers. |
| <b>Internet requirements</b> | Internet connectivity will be required to access the system.                                                 |
| <b>Organisation fit</b>      | The solution supports CUT's goal of moving from paper-based processes towards a paperless institution.       |
| <b>Training</b>              | Staff and administrators may require basic system training.                                                  |

|                |                                                                                 |
|----------------|---------------------------------------------------------------------------------|
| <b>Privacy</b> | Access to sensitive student information must be restricted to authorised users. |
|----------------|---------------------------------------------------------------------------------|

The stakeholder specifically stated that the system should support CUT's move towards becoming a paperless institution by 2030.

### 14.3 Economic feasibility

The economic feasibility assessment focuses on whether the group has access to the resources required to develop and demonstrate the prototype.

| Resource                      | Availability | Action Required                                                                                                                    |
|-------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Internet/Data</b>          | Available    | Team members need ensure they have sufficient personal data for development and meetings. Unless utilize the university free data. |
| <b>Transport</b>              | Available    | Arrange transport for stakeholder meetings when required.                                                                          |
| <b>Development computers</b>  | Available    | Use available computers for development.                                                                                           |
| <b>Test devices</b>           | Available    | Use smartphones and computers for testing.                                                                                         |
| <b>Development software</b>   | Available    | Use required/free development tools.                                                                                               |
| <b>Supabase</b>               | Available    | Use Supabase for the shared backend/database.                                                                                      |
| <b>Presentation resources</b> | Available    | Use available university/personal resources.                                                                                       |
| <b>Other resources</b>        | Available    | Use existing group resources where possible.                                                                                       |

### Economic Feasibility Decision

#### Decision: Feasible

The project is economically feasible for the proposed prototype because the group can use existing development resources and free/open-source tools and services. The project constraints specify that the budget is limited to free/open-source tools and services.

### 14.4 Schedule feasibility

#### Evaluate Schedule Feasibility

| Question                                           | Group finding                                                                                |
|----------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Is enough time available for system design?</b> | <b>Yes.</b> The original time of 3 weeks allocation in the beginning was enough, but we fear |

|                                                                         |                                                                                                                                                       |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         | that the time is shorten time will be tight. Since it reduced to 2 weeks                                                                              |
| <b>Is enough time available for Flutter development?</b>                | Yes ,with focus, with the allocated roles and everybody doing and completing their tasks it will be developed in time                                 |
| <b>Is enough time available for ASP.NET development?</b>                | <b>Yes, with focus.</b> We have 3 weeks of development                                                                                                |
| <b>Is enough time available for Supabase integration?</b>               | Yes. Supabase setup, authentication, database design, and real time communication can be completed within 2-3 weeks, integrated alongside development |
| <b>Is enough time available for testing and corrections?</b>            | <b>Yes.</b> Last week of development, allocated for manual testing, stakeholder feedback, and corrections. 1 week is tight but manageable for MVP.    |
| <b>Is enough time available for stakeholder feedback?</b>               | <b>Yes.</b> Communication with the stakeholder is simple and quick, a meeting will be scheduled ahead of time                                         |
| <b>Are too many functions currently included?</b>                       | No we actually want to add more                                                                                                                       |
| <b>Which functions may need to be simplified, postponed or removed?</b> | No, the feature make the project                                                                                                                      |

### Schedule Feasibility Decision

The project is achievable within the 8 weeks semester **provided we prioritise core features**. Development time has been reduced (system design now 2 weeks instead of 3), which is tight but manageable with clear role allocation. Adding more features increases risk of incomplete delivery. We will deliver MVP features first and treat additional features as stretch goals.

## 14.5 Legal feasibility

### Evaluate legal Feasibility

| Question                                                | Group finding                                                                                                                                                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Will the prototype collect personal information?</b> | Yes. Student name, email, student number, mood logs, journal entries, appointment history, messages                                                                                     |
| <b>Is all the information necessary?</b>                | Yes. All data collected serves a functional purpose: identification (email/student number), mental health tracking (mood/journal), appointment management, and communication (messages) |
| <b>Who should be allowed to access the information?</b> | Students, Therapist, Admin for data verification and management                                                                                                                         |

|                                                                       |                                                                                                      |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Will test data be used during development?</b>                     | <b>Yes.</b> All development and testing will use anonymised data. No real student data will be used. |
| <b>How will passwords and private keys be protected?</b>              | Passwords hashed using Supabase Auth                                                                 |
| <b>Does the project involve confidential stakeholder information?</b> | No is does not. The Wellness Center is providing general process information only.                   |
| <b>Is private stakeholder information being kept out of GitHub?</b>   | Yes                                                                                                  |

### Legal Feasibility Decision

The project is legally feasible. All data collected is necessary and consensual. Only test data will be used during development. Passwords are hashed using Supabase Authentication. No confidential stakeholder information is involved. Sensitive credentials will be excluded from GitHub. POPIA compliance will be maintained (consent, access rights, data minimisation).

### 14.5 Ethical Feasibility

#### Ethical Feasibility

| Question                                                             | Group finding                                                                                                                                                                 |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Does the stakeholder understand the purpose of the project?</b>   | <b>Yes.</b> Stakeholder has been clearly informed it is an <b>educational prototype</b> . They agreed to participate.                                                         |
| <b>Is unnecessary personal information being collected?</b>          | <b>No.</b> Only necessary information: email (verification), student number (identity), mood/journal/messages (core functionality). No sensitive health data without consent. |
| <b>Could the proposed solution negatively affect intended users?</b> | No, the risk is low, it provides                                                                                                                                              |
| <b>Is information being used only for its intended purpose?</b>      | <b>Yes.</b> Informed consent at registration. Opt-in/out for data sharing. Right to access, correct, and delete data (POPIA). The data is anonymous for analysis.             |

#### Summary of evaluation

The project is ethically feasible. Stakeholder understands it is an educational prototype and has agreed to participate. Only necessary data is collected. Informed consent is obtained at registration with opt-in/out options for data sharing. Users have rights to access, correct, and delete their data. The risk of negative impact is low as the app provides supportive tools.

## 14.6 Feasibility conclusion

| Feasibility Area   | Decision              | Main Reason                                                                                                                                | Change Required                                                     |
|--------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Technical</b>   | Feasible with changes | Some advanced features require additional technical skills and development time                                                            | conduct early proof-of-concept                                      |
| <b>Operational</b> | Feasible              | The system directly addresses Wellness Centre's current problems and fits the intended environment                                         | Schedule regular stakeholder feedback meetings                      |
| <b>Economic</b>    | Feasible              | The prototype can be developed using available resources: Supabase free tier, Flutter, ASP.NET open-source tools. No major costs involved. | None                                                                |
| <b>Schedule</b>    | Feasible with changes | Timeline is tight (design reduced to 2 weeks), but manageable with clear role allocation and scope prioritisation                          | Prioritise MVP features; treat additional features as stretch goals |
| <b>Legal</b>       | Feasible              | POPIA compliance planned. Only fictional/test data used. No sensitive info on GitHub. Passwords hashed.                                    | Ensure consent forms and data deletion functionality are included   |
| <b>Ethical</b>     | Feasible              | Stakeholder informed. Data minimisation. Informed consent with opt-in/out. Low risk to users.                                              | Test emergency features thoroughly                                  |

## Overall Feasibility Decision

The proposed Student Wellness System is **feasible with some changes**. It addresses the Wellness Centre's manual processes while adding wellness features like mood tracking, journaling, messaging, and student verification.

Core MVP features (appointment booking, mood tracking, journaling, secure messaging, student verification, emergency panic button, and admin dashboard) will be delivered within the 9-week semester. Additional features will be developed only if time permits.

We will prioritise Must Have requirements and postpone lower-priority features where necessary. The project is technically, operationally, economically, legally, and ethically sound. All development and testing will use fictional/anonymised data only.

**Decision: Feasible with changes**

## Project Risk Register

| Score Scale: | Level  |
|--------------|--------|
| >=3          | Low    |
| >=6          | Medium |
| >=9          | High   |

| Risk ID | Risk Statement                                                                                                                                                                                          | Category                       | Likelihood (1-3) | Impact (1-3) | Score | Level  | Mitigation Notes                                                                                                                                        | Contingency                                                                                                                   | Risk Owner | Status |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------|--------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| R01     | The project team may not fully document the stakeholder's current Wellness Centre process and needs, which could result in incomplete or inaccurate requirements.                                       | Requirements                   | 2                | 3            | 6     | Medium | Compare the stakeholder interview findings with the current Wellness Centre process and review the findings as a group before finalizing requirements.  | Re-engage with the stakeholder to clarify missing information and update the requirements before the final approval.          | LD Mokheti | Open   |
| R02     | The project team may incorrectly define the system scope. Resulting in features being included that are not required or feasible within the project timeframe.                                          | Scope                          | 3                | 3            | 9     | High   | Review the proposed features against the stakeholder needs, feasibility findings and the available project timeframe before approving the scope.        | Remove or postpone unnecessary features and revise the scope to focus on the required and feasible functionality.             | BF Motseki | Open   |
| R03     | Development tasks may take longer than planned, leaving insufficient time for testing, corrections and final demonstrations.                                                                            | Requirements and Documentation | 2                | 3            | 6     | Medium | Review the requirements against the stakeholder findings, the current process and project scope before approving the SRS.                               | Conduct an additional requirements review and update the SRS before moving into the detailed design.                          | LD Mokheti | Open   |
| R04     | The feasibility analysis may overlook important technical, resource or time constraints, which could result in requirements being selected that the team cannot realistically implement.                | Feasibility                    | 2                | 3            | 6     | Medium | Evaluate the technical, resource and time constraints before finalising the requirements and proposed features.                                         | Reassess the affected requirements and modify or remove requirements that cannot realistically be implemented.                | A Jara     | Open   |
| R05     | The project team may underestimate the technical, economic, operational, schedule or legal feasibility of the proposed system, which could result in an unrealistic project baseline.                   | Feasibility                    | 2                | 3            | 6     | Medium | Evaluate the technical, economic, operational, schedule and legal feasibility before establishing the project baseline.                                 | Reassess the feasibility findings and adjust the scope, requirements or schedule to establish a realistic baseline.           | A Jara     | Open   |
| R06     | The project team may fail to identify important project risks, constraints and assumptions. Resulting in potential problems not being considered in the project plan.                                   | Risk Management                | 2                | 3            | 6     | Medium | Conduct a structured Phase 1 review covering project risks, constraints and assumptions before approving the baseline.                                  | Conduct another risk review and update the risk register and project plan with the missing risks, constraints or assumptions. | A Jara     | Open   |
| R07     | The system architecture and Supabase database design may not fully support the approved requirements, user roles and shared information required by both applications.                                  | Architecture/Data base         | 2                | 3            | 6     | Medium | Review the proposed architecture and Supabase database design against the approved requirements and user roles before development.                      | Review and modify the architecture or database design before development progresses further.                                  | B Mbinga   | Open   |
| R08     | The integration between the Flutter application, ASP.NET web application and Supabase may not function as required, which could result in information not being shared correctly between the platforms. | Integration/ Tesing            | 3                | 3            | 9     | High   | Define integration requirements early and perform integration checks to confirm that Flutter and ASP.NET can use the shared Supabase backend correctly. | Investigate the integration problem, prioritise the core data sharing functionality and correct the affected components.      | TV Thabisi | Open   |
| R09     | The Flutter mobile application may not implement the approved requirements correctly, which could result in missing or incorrectly functioning mobile features.                                         | Development /Flutter           | 2                | 3            | 6     | Medium | The developers should develop the Flutter application against the approved SRS and review each implemented feature against the agreed requirements.     | Review the affected Flutter functionality against the approved requirements and correct or complete the affected features.    | ML Mwenda  | Open   |

## Feasibility and Risk Register Review

| Finding                 | Does the project need to change ? | What change is required ?                                                                                                                     |
|-------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Technical Feasibility   | No                                |                                                                                                                                               |
| Operational Feasibility | Yes                               | Schedule regular stakeholder feedback meetings throughout development.                                                                        |
| Economic Feasibility    | Yes                               | Utilise university free data where available. No major cost changes required.                                                                 |
| Schedule Feasibility    | yes                               | Prioritise MVP features, Postpone advanced analytics to stretch goals. Remove features that cannot be completed within 9 weeks.               |
| Legal Feasibility       | yes                               | Include consent forms, data deletion, and data access functionality to ensure POPIA compliance.                                               |
| Ethical Feasibility     | yes                               | Thoroughly test emergency panic button and SOS features. If testing reveals issues, simplify or provide fallback (display crisis line number) |
| High Project risks      | yes                               | Remove or postpone unnecessary features, review each feature against requirements                                                             |

### 3. Refined the Project Scope

| Function Requirement                | Decision | Reason                                         | Stakeholder Informed ? |
|-------------------------------------|----------|------------------------------------------------|------------------------|
| Register and login                  | Keep     | Essential for system access                    | Yes                    |
| Digital Wellness Centre Intake Form | Keep     | Core requirement for digitising manual process | Yes                    |
| Next-of-kin information             | Keep     | Required for emergency contact                 | Yes                    |
| Purpose of visit selection          | Keep     | Essential for triage and reporting             | Yes                    |

|                                     |          |                                                            |         |
|-------------------------------------|----------|------------------------------------------------------------|---------|
| <b>Student mood tracking</b>        | Keep     | Core wellness feature                                      | Yes     |
| <b>Student journaling</b>           | Keep     | Core wellness feature                                      | Yes     |
| <b>Student emotional check-ins</b>  | Keep     | Core wellness feature                                      | Yes     |
| <b>Secure messaging</b>             | Keep     | Stakeholder requirement for student-provider communication | Yes     |
| <b>Emergency SOS / Panic button</b> | Keep     | Critical safety feature                                    | Yes     |
| <b>Advanced reporting</b>           | Simplify | Basic reports only; detailed analytics postponed           | Yes     |
| <b>SI/Tutor integration</b>         | Postpone | Requires additional coordination and scope expansion       | Pending |

#### 4. Updated the Requirements

| Requirement ID | Original requirement                                   | Change made                             | Reason for change                                    |
|----------------|--------------------------------------------------------|-----------------------------------------|------------------------------------------------------|
| <b>FR14</b>    | The student should be able to use medication reminders | Simplified to basic push notifications  | Reduces development complexity                       |
| <b>FR15</b>    | The system should provide streaks and achievements     | pending                                 | Additional development time required                 |
| <b>FR16</b>    | The system should provide additional reporting         | Simplified to basic appointment reports | Advanced analytics postponed due to time constraints |